METHANOL CONVERTER VER 1

Operator Training Simulator · R-201 quench converter and synthesis loop

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DOCUMENT REGISTER

Documents, information flow, engineering queries and progress of the R-201 loop

DOCUMENT NO. : MEOH-V1-PM-DOC-001

DOCUMENT TITLE : Document Register

PROJECT : Methanol Converter Ver 1 · OTS

STATUS : Rev 0 · Issued for teaching

Teaching document generated from the Methanol Converter Ver 1 simulator model; numbers come from the model and the OTS tabs. Not for construction.

Rev.	Date	Revision status	Originator	Checker	Approver
0	27-Oct-26	Issued for teaching	A. Soottitantawat		

1 Purpose and scope

This register lists every document of the Methanol Converter Ver 1 synthesis loop (R-201, C-201, E-201…E-204, V-201): the EPC-style documents generated from the simulator model for the Engineering (EPC) tabs, the drawings and studies that already existed (PFD-201, the EPC P&IDs MEOH-PID-001/002, the C&E diagram, the SRS and the HAZOP report) and this register. For each document it gives the number, title, discipline, revision and status, the page count read from the issued PDF, the file name, the tab of the OTS page that teaches it, and what it is fed by and what it feeds. It also carries the engineering-query register of the EPC tabs and a progress measurement of the document set.

All documents are issued for teaching. There is no client, contractor or vendor behind them; the numbers come from the simulator model and the OTS tabs. The PDFs sit next to the OTS page and are linked from it by bare file name.

1.1 Document numbering

Unit documents are numbered **MEOH-V1-<discipline>-<type>-<nnn>**: MEOH-V1 = the unit (methanol converter, version 1 of the simulator); discipline code (two letters); document-type code; serial number within the type. Datasheets carry one serial per item (MEOH-V1-ME-DS-001 = R-201 … -007 = C-201). The two P&ID sheets and PFD-201 were drawn before the document set and keep their drawing numbers; the layout MEOH-V1-LAY-001 carries no discipline code (engineering query EQ-ej-01). Visual plant 3D (section 3) is a tab, not a document.

Code	Discipline	Type codes used here
PM	Project management	COM, DOC, PKG
PE	Process	HMB, LSL, RLB
ME	Mechanical	DS, EQL
PL	Piping & layout	MEOH-V1-LAY-001 (no discipline code in the number)
IN	Instrumentation & control	CE, INL, SRS
EE	Electrical	HAC, LL
SA	Safety	HZP

1.2 Revision codes

Code	Meaning (typical EPC practice)	In this document set
A	Issued for review (IFR): first issue, the other disciplines and the client comment	MEOH-PID-001/002 ("issued for OTS development (review)"), the HAZOP report
B	Issued for approval (IFA): comments incorporated, for the client’s approval	none
0	Issued for construction (IFC) or for purchase; later revisions 1, 2 … with clouded changes	Rev 0 = "Issued for teaching": the EPC documents, PFD-201, this register
R0	Not a standard code: written by two documents for their first issue	C&E MEOH-V1-IN-CE-001, SRS MEOH-V1-IN-SRS-001 (EQ-ej-02)

A document is re-issued with a new revision when an engineering query against it is accepted; the query register (section 4) is therefore the to-do list of the next revision.

2 Document register

#	Document no.	Title	Disc.	Rev · status	Date	Pp.	File	Taught in tab	Fed by	Feeds
1	**MEOH-V1-PM-DOC-001**	Document Register	PM	0 · Issued for teaching	27-Oct-26	17	MEOH-V1-PM-DOC-001.pdf	Project & deliverables	every document, epc/queries	project control (progress, query close-out)
2	**PFD-201**	Process flow diagram · R-201 synthesis loop Process flow diagram	PE	0 · Issued for teaching (OTS HAZID study drawing)	29-Sep-26	1	pfd-201.pdf	PFD	Simulator model	PE-HMB-001, MEOH-PID-001/002, PM-DOC-001
3	**MEOH-PID-001**	P&ID · Methanol converter R-201, interchanger E-201, E-204 P&ID (EPC sheet 1 of 2, alternative version)	PE	A · Issued for OTS development (review)	25-SEP-26	1	MEOH-PID-001.pdf	P&ID	PFD-201	SA-HZP-001, PE-LSL-001, ME-EQL-001, PE-RLB-001, IN-INL-001, EE-HAC-001, LAY-001, PM-COM-001, PM-DOC-001
4	**MEOH-PID-002**	P&ID · Synthesis loop: C-201, E-202, E-203, separator V-201 P&ID (EPC sheet 2 of 2, alternative version)	PE	A · Issued for OTS development (review)	25-SEP-26	1	MEOH-PID-002.pdf	P&ID	PFD-201	SA-HZP-001, PE-LSL-001, ME-EQL-001, PE-RLB-001, IN-INL-001, EE-HAC-001, LAY-001, PM-COM-001, PM-DOC-001
5	**MEOH-V1-PE-HMB-001**	Heat and Material Balance	PE	0 · Issued for teaching	27-Oct-26	12	MEOH-V1-PE-HMB-001.pdf	Process engineering	Simulator model, PFD-201	PE-LSL-001, ME-EQL-001, ME-DS-001…007, PE-RLB-001, IN-INL-001, EE-LL-001, EE-HAC-001, PM-DOC-001
6	**MEOH-V1-PE-LSL-001**	Line Sizing List	PE	0 · Issued for teaching	27-Oct-26	7	MEOH-V1-PE-LSL-001.pdf	Process engineering	PE-HMB-001, MEOH-PID-001/002	LAY-001, PM-DOC-001
7	**MEOH-V1-PE-RLB-001**	Relief Load Basis	PE	0 · Issued for teaching	27-Oct-26	7	MEOH-V1-PE-RLB-001.pdf	Relief & vent	PE-HMB-001, MEOH-PID-001/002, ME-EQL-001, SA-HZP-001	PM-PKG-001, PM-DOC-001
8	**MEOH-V1-ME-EQL-001**	Equipment List	ME	0 · Issued for teaching	27-Oct-26	6	MEOH-V1-ME-EQL-001.pdf	Mechanical	PE-HMB-001, MEOH-PID-001/002	ME-DS-001…007, PE-RLB-001, PM-PKG-001, LAY-001, PM-COM-001, PM-DOC-001

#	Document no.	Title	Disc.	Rev · status	Date	Pp.	File	Taught in tab	Fed by	Feeds
9	**MEOH-V1-ME-DS-001**	Datasheet R-201 Equipment datasheet	ME	0 · Issued for teaching	27-Oct-26	4	MEOH-V1-ME-DS-001.pdf	Mechanical	ME-EQL-001, PE-HMB-001	EE-LL-001, PM-PKG-001, PM-DOC-001
10	**MEOH-V1-ME-DS-002**	Datasheet V-201 Equipment datasheet	ME	0 · Issued for teaching	27-Oct-26	3	MEOH-V1-ME-DS-002.pdf	Mechanical	ME-EQL-001, PE-HMB-001	EE-LL-001, PM-PKG-001, PM-DOC-001
11	**MEOH-V1-ME-DS-003**	Datasheet E-201 Equipment datasheet	ME	0 · Issued for teaching	27-Oct-26	2	MEOH-V1-ME-DS-003.pdf	Mechanical	ME-EQL-001, PE-HMB-001	EE-LL-001, PM-PKG-001, PM-DOC-001
12	**MEOH-V1-ME-DS-004**	Datasheet E-202 Equipment datasheet	ME	0 · Issued for teaching	27-Oct-26	2	MEOH-V1-ME-DS-004.pdf	Mechanical	ME-EQL-001, PE-HMB-001	EE-LL-001, PM-PKG-001, PM-DOC-001
13	**MEOH-V1-ME-DS-005**	Datasheet E-203 Equipment datasheet	ME	0 · Issued for teaching	27-Oct-26	2	MEOH-V1-ME-DS-005.pdf	Mechanical	ME-EQL-001, PE-HMB-001	EE-LL-001, PM-PKG-001, PM-DOC-001
14	**MEOH-V1-ME-DS-006**	Datasheet E-204 Equipment datasheet	ME	0 · Issued for teaching	27-Oct-26	2	MEOH-V1-ME-DS-006.pdf	Mechanical	ME-EQL-001, PE-HMB-001	EE-LL-001, PM-PKG-001, PM-DOC-001
15	**MEOH-V1-ME-DS-007**	Datasheet C-201 Equipment datasheet	ME	0 · Issued for teaching	27-Oct-26	2	MEOH-V1-ME-DS-007.pdf	Mechanical	ME-EQL-001, PE-HMB-001	EE-LL-001, PM-PKG-001, PM-DOC-001
16	**MEOH-V1-LAY-001**	Plot Plan, Elevations and Piping Isometrics	PL	0 · Issued for teaching	27-Oct-26	34	MEOH-V1-LAY-001.pdf	Layout & isometrics	ME-EQL-001, MEOH-PID-001/002, PE-LSL-001, EE-HAC-001	Visual plant 3D tab (3-D model, field rounds), PM-DOC-001
17	**MEOH-V1-IN-INL-001**	Instrument List	IN	0 · Issued for teaching	27-Oct-26	8	MEOH-V1-IN-INL-001.pdf	Instrumentation & control	MEOH-PID-001/002, PE-HMB-001, IN-CE-001, IN-SRS-001	EE-LL-001, PM-PKG-001, PM-COM-001, PM-DOC-001
18	**MEOH-V1-IN-CE-001**	Cause and effect diagram	IN	R0 · Issued for teaching	06-Oct-26	2	MEOH-V1-IN-CE-001.pdf	C&E matrix	SA-HZP-001, IN-SRS-001	IN-INL-001, PM-COM-001, PM-DOC-001
19	**MEOH-V1-IN-SRS-001**	Safety requirements specification of the SIFs Safety requirements specification	IN	R0 · Issued for teaching (tab and CSV; no PDF issued)	—	—	— (tab only)	SRS of SIF	SA-HZP-001	IN-CE-001, IN-INL-001, PM-PKG-001, PM-DOC-001
20	**MEOH-V1-EE-LL-001**	Electrical Load List	EE	0 · Issued for teaching	27-Oct-26	5	MEOH-V1-EE-LL-001.pdf	Electrical	PE-HMB-001, ME-DS-001…007, IN-INL-001	PM-PKG-001, PM-DOC-001
21	**MEOH-V1-EE-HAC-001**	Hazardous-Area Classification	EE	0 · Issued for teaching	27-Oct-26	4	MEOH-V1-EE-HAC-001.pdf	Electrical	MEOH-PID-001/002, PE-HMB-001	PM-PKG-001, LAY-001, PM-DOC-001

#	Document no.	Title	Disc.	Rev · status	Date	Pp.	File	Taught in tab	Fed by	Feeds
22	**MEOH-V1-SA-HZP-001**	HAZOP report (IEC 61882) HAZOP report	SA	A · Issued for teaching	06-Oct-26	59	MEOH-V1-SA-HZP-001.pdf	HAZOP	MEOH-PID-001/002	PE-RLB-001, IN-SRS-001, IN-CE-001, PM-DOC-001
23	**MEOH-V1-PM-PKG-001**	Procurement Package List	PM	0 · Issued for teaching	27-Oct-26	10	MEOH-V1-PM-PKG-001.pdf	Procurement to commissioning	ME-EQL-001, ME-DS-001…007, PE-RLB-001, IN-INL-001, EE-LL-001, EE-HAC-001, IN-SRS-001	PM-COM-001, PM-DOC-001
24	**MEOH-V1-PM-COM-001**	Pre-commissioning and Commissioning Plan	PM	0 · Issued for teaching	27-Oct-26	13	MEOH-V1-PM-COM-001.pdf	Procurement to commissioning	PM-PKG-001, IN-CE-001, ME-EQL-001, IN-INL-001, MEOH-PID-001/002	PM-DOC-001

Pages are counted from the issued PDF files (pypdf). 24 documents, 204 pages in 23 PDF files. Discipline codes as in section 1.1; the P&IDs and PFD-201 are counted under PE. In the last two columns the prefix MEOH-V1- is left out.

3 Deliverable tree and information flow

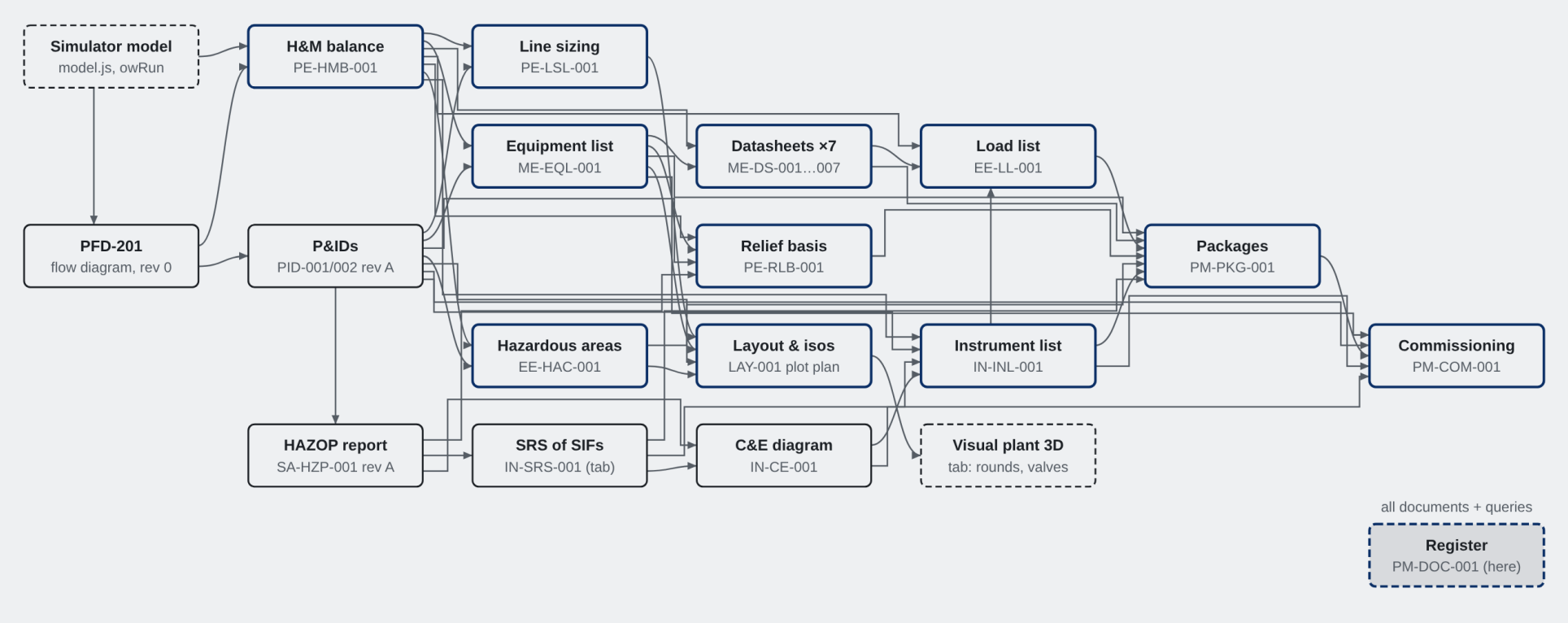


Figure 1. Information flow between the documents (from the Project & deliverables tab). An arrow means the receiving document takes data from the sending one.

Each arrow below is one hand-over. "Cited" is checked in the issued PDF of the receiving document: ✓ when its text quotes the sender's number, "not by number" when it does not (the data may still come from it, e.g. through the model or the tab); "—" when the receiver has no PDF (the SRS) or the sender is the model.

From	To	What passes	Cited
Simulator model (model.js, owRun)	PFD-201 (process flow diagram)	design point, stream data	—
Simulator model (model.js, owRun)	Heat & material balance (PE-HMB-001)	design and off-design runs (owRun)	—
PFD-201 (process flow diagram)	Heat & material balance (PE-HMB-001)	stream numbers 1–12	✓
PFD-201 (process flow diagram)	P&IDs (MEOH-PID-001 / 002 rev A)	process scheme, tags	not by number
P&IDs (MEOH-PID-001 / 002 rev A)	HAZOP report (SA-HZP-001 rev A)	nodes and design intents	✓
Heat & material balance (PE-HMB-001)	Line sizing list (PE-LSL-001)	flows and densities per case	✓
P&IDs (MEOH-PID-001 / 002 rev A)	Line sizing list (PE-LSL-001)	line numbers and drawn sizes	✓
Heat & material balance (PE-HMB-001)	Equipment list (ME-EQL-001)	operating P, T and duties	✓
P&IDs (MEOH-PID-001 / 002 rev A)	Equipment list (ME-EQL-001)	equipment tags	✓

From	To	What passes	Cited
Equipment list (ME-EQL-001)	Datasheets ×7 (ME-DS-001…007)	design P / T, materials	✓
Heat & material balance (PE-HMB-001)	Datasheets ×7 (ME-DS-001…007)	duties and flows	not by number
Heat & material balance (PE-HMB-001)	Relief load basis (PE-RLB-001)	relief loads	✓
P&IDs (MEOH-PID-001 / 002 rev A)	Relief load basis (PE-RLB-001)	PSV-201, HV-205	✓
Equipment list (ME-EQL-001)	Relief load basis (PE-RLB-001)	design pressures (set ≤ MAWP)	✓
HAZOP report (SA-HZP-001 rev A)	Relief load basis (PE-RLB-001)	recommendations R03, R15	not by number
HAZOP report (SA-HZP-001 rev A)	SRS of the SIFs (IN-SRS-001 (tab))	SIF demands	—
HAZOP report (SA-HZP-001 rev A)	C&E diagram (IN-CE-001)	trips and actions	not by number
SRS of the SIFs (IN-SRS-001 (tab))	C&E diagram (IN-CE-001)	SIF logic	not by number
P&IDs (MEOH-PID-001 / 002 rev A)	Instrument list (IN-INL-001)	instrument tags	✓
Heat & material balance (PE-HMB-001)	Instrument list (IN-INL-001)	ranges, normal values	not by number
C&E diagram (IN-CE-001)	Instrument list (IN-INL-001)	alarm and trip set points	✓
SRS of the SIFs (IN-SRS-001 (tab))	Instrument list (IN-INL-001)	SIS devices	✓
Heat & material balance (PE-HMB-001)	Electrical load list (EE-LL-001)	C-201 power	not by number
Datasheets ×7 (ME-DS-001…007)	Electrical load list (EE-LL-001)	C-201 driver	not by number
Instrument list (IN-INL-001)	Electrical load list (EE-LL-001)	instrument and analyser power	✓
P&IDs (MEOH-PID-001 / 002 rev A)	Hazardous areas (EE-HAC-001)	release sources	✓
Heat & material balance (PE-HMB-001)	Hazardous areas (EE-HAC-001)	gas compositions	not by number
Equipment list (ME-EQL-001)	Procurement packages (PM-PKG-001)	scope and design data	✓
Datasheets ×7 (ME-DS-001…007)	Procurement packages (PM-PKG-001)	datasheets for the requisitions	✓
Relief load basis (PE-RLB-001)	Procurement packages (PM-PKG-001)	PSV sizes	✓
Instrument list (IN-INL-001)	Procurement packages (PM-PKG-001)	instrument scope	✓
Electrical load list (EE-LL-001)	Procurement packages (PM-PKG-001)	motor and switchgear	✓
Hazardous areas (EE-HAC-001)	Procurement packages (PM-PKG-001)	Ex ratings	✓
SRS of the SIFs (IN-SRS-001 (tab))	Procurement packages (PM-PKG-001)	SIS package	✓
Equipment list (ME-EQL-001)	Layout & isometrics (LAY-001)	equipment sizes, nozzles	✓
P&IDs (MEOH-PID-001 / 002 rev A)	Layout & isometrics (LAY-001)	lines, valves, instruments	✓
Line sizing list (PE-LSL-001)	Layout & isometrics (LAY-001)	line list: numbers and sizes	✓
Hazardous areas (EE-HAC-001)	Layout & isometrics (LAY-001)	zones on the plot plan	✓
Layout & isometrics (LAY-001)	Visual plant 3D (3-D model, field rounds (tab))	layout.json: positions, levels, gauges, hand valves	—
Procurement packages (PM-PKG-001)	Commissioning plan (PM-COM-001)	FAT and delivery	not by number
C&E diagram (IN-CE-001)	Commissioning plan (PM-COM-001)	FAT / SAT test sheets	✓
Equipment list (ME-EQL-001)	Commissioning plan (PM-COM-001)	test pressures	✓
Instrument list (IN-INL-001)	Commissioning plan (PM-COM-001)	loop checks	✓
P&IDs (MEOH-PID-001 / 002 rev A)	Commissioning plan (PM-COM-001)	test packs, line-ups	✓

30 of 40 checkable hand-overs are cited by number. Not cited: PFD-201 → P&IDs; Heat & material balance → Datasheets ×7; HAZOP report → Relief load basis; HAZOP report → C&E diagram; SRS of the SIFs → C&E diagram; Heat & material balance → Instrument list; Heat & material balance → Electrical load list; Datasheets ×7 → Electrical load list; Heat & material balance → Hazardous areas; Procurement packages → Commissioning plan.

4 Engineering-query register

Every disagreement found while writing the documents (between two documents, or a document and the model, an alarm limit or a trip) is recorded as an engineering query (EQ) by the tab that found it, addressed to the owner of the other document, and closed by an answer or a revision. Status "open" = no answer yet; "answered" = the answer is recorded (the receiving document still has to be revised where the answer says so). EQ-ej-01…04 are the register's own document-control queries.

Raised in (tab)	Open	Answered	Total
Process engineering	6	1	7
Mechanical	5	0	5
Relief & vent	7	0	7
Instrumentation & control	8	0	8
Electrical	5	0	5
Procurement to commissioning	6	1	7
Layout & isometrics	5	0	5
Project & deliverables	4	0	4
All	**46**	**2**	**48**

Addressed to (a query can name several)	Queries	Open
P&IDs	24	24
Equipment list / datasheets	5	3
Relief load basis	3	2
Simulator model	7	7
HMB / LSL / PFD	6	6
HAZOP	6	6
C&E / SRS	10	10
Other (SOP, front end, downstream)	4	4
INL / electrical	1	1
Structural / platforms	1	1
Layout	1	1

No.	Raised in	From	To	Issue	Proposed	Status
EQ-ep-01	Process engineering	MEOH-V1-PE-LSL-001	MEOH-PID-001 / MEOH-PID-002	At the rated flow (the larger of normal × 1.10 and the highest-flow case) 7 of the 10 numbered gas lines exceed the 20 m/s basis of P&ID note 2 with the pipe walls for the design pressure zones of MEOH-V1-ME-EQL-001: DN750-PG-201 26.32 m/s (needs DN900); DN350-PG-232 25.79 m/s (needs DN400); DN350-PG-242 24.85 m/s (needs DN400); DN600-PG-213 31.72 m/s (needs DN800); DN900-PG-251 25.7 m/s (needs DN1050); DN300-SG-202 22.76 m/s (needs DN350); DN750-PG-202 22.64 m/s (needs DN800). ρv² and ΔP per 100 m are within the typical criteria on all of them except DN300-SG-202 (ρv² 16851).	Either increase these lines to the sizes listed (with piping to confirm wall and ID), or revise P&ID note 2 to a ρv²-based criterion for the high-pressure loop gas (typical ρv² ≤ 15 000 kg/(m·s²)) and keep the drawn sizes where ρv² is met; check the loop ΔP budget against C-201 either way.	open
EQ-ep-02	Process engineering	MEOH-V1-PE-LSL-001	MEOH-PID-002	DN150-PU-203 carries the purge (105.1 kNm³/h normal, 37875 kg/h rated). Upstream of PV-203 at 75 bar g it runs at 31.6 m/s with ρv² 19748 kg/(m·s²) and 1.027 bar/100 m, outside all three typical criteria (DN200 meets them); the line number continues downstream of PV-203 to the reformer fuel header, where at an assumed 4 bar g the same flow would run at about 557 m/s (Mach 0.72). The fuel-header pressure is not in the model or on the P&ID.	Increase the upstream section to DN200; break the line number at the PV-203 spec break; size the downstream line for the fuel-header pressure (DN700 at 4 bar g with the typical 30 m/s limit) and confirm the header pressure; check PV-203 for noise.	open
No.	**Raised in**	**From**	**To**	**Issue**	**Proposed**	**Status**
EQ-ep-03	Process engineering	MEOH-V1-PE-HMB-001	MEOH-V1-PE-RLB-001 /	PSV-201 (set 88 bar g) is on V-201, the low-pressure end of the loop. With C-201 running at design circulation and	Set the design pressure of the C-201 discharge side at PSV set + loop ΔP (≈ 93 bar g, before	answered by MEOH-V1-ME-EQL-001

No.	Raised in	From	To	Issue	Proposed	Status
			MEOH-V1-ME-EQL-001	V-201 at the PSV set, the C-201 discharge side (E-202 cold side, R-201, E-201) sits at about 92.8 bar g, above the 88 bar g that the V-201 design pressure is based on.	accumulation) or add relief on the C-201 discharge; confirm in the relief load basis and the equipment list.	(zone B 100 bar g)
EQ-ep-04	Process engineering	MEOH-V1-PE-HMB-001	MEOH-PID-001 / -002 (design-point table)	The P&ID design-point table gives outlet CH3OH 2.1 / 3.6 / 4.4 / 4.8 % for beds 1–4; the model at design gives 1.96 / 3.59 / 4.29 / 4.78 %, i.e. 2.0 / 3.6 / 4.3 / 4.8 % to the table's one decimal: bed 1 and bed 3 differ. Bed temperatures agree within 0.5 °C.	Revise the P&ID design-point table CH3OH column to 2.0 / 3.6 / 4.3 / 4.8 % at the next revision (the HMB governs).	open
EQ-ep-05	Process engineering	MEOH-V1-PE-HMB-001	MEOH-PID-002 note 1	Note 1 labels H2 81.6 / CH4 11.5 / CO2 2.5 / CO 2.3 / N2 1.7 / CH3OH 0.36 mol % as the loop gas "at C-201 suction"; that is the V-201 vapour (stream 10: H2 81.6, CH4 11.5). The make-up joins upstream of C-201, so the suction gas is stream 3: H2 80.0 / CH4 9.9 / CO2 3.4 / CO 4.9 / N2 1.5 / CH3OH 0.29 mol %.	Relabel note 1 as "V-201 vapour / recycle gas" or quote the stream 3 composition for C-201 suction.	open
EQ-ep-06	Process engineering	MEOH-V1-PE-LSL-001	MEOH-PID-001 / MEOH-PID-002	8 process lines on the P&IDs have no line number (e.g. the E-201 → E-202 → E-203 → V-201 effluent train, the C-201 suction after the make-up tie-in, the TV-211 bypass, the PSV-201 and HV-205 flare lines), so the line sizing list cannot cover them.	Assign line numbers (and sizes) on the next P&ID revision; size them from the HMB streams given in MEOH-V1-PE-LSL-001 section 5.	open
EQ-ep-07	Process engineering	MEOH-V1-PE-HMB-001	PFD tab equipment list / OTS model (E-203)	The PFD tab lists E-203 as "≈ 84 MW sensible + condensing". The HMB gives 83.76 MW sensible (the model's E-203 heat, which also sets the cooling-water return TI-254) plus 49.18 MW to condense the crude methanol (latent heats assumed), 132.94 MW in all. The model leaves the latent part out, so TI-254 reads 41.31 °C where the full duty would give about 46.8 °C at the same cooling-water flow.	Quote E-203 as ≈ 84 MW sensible + ≈ 49 MW condensing (≈ 133 MW) on the PFD tab; size E-203 and the cooling water on the full duty; record the sensible-only TI-254 as a known model simplification.	open
EQ-em-01	Mechanical	MEOH-V1-ME-EQL-001	MEOH-PID-001 / MEOH-PID-002	The P&IDs give no design pressure or temperature for any equipment (V-201 shows only its operating 75 bar g / 40 °C). The equipment list sets two pressure zones: zone A (V-201, C-201 suction) 88 bar g = PSV-201 set, and zone B (C-201 discharge through R-201 and the exchangers to the V-201 inlet) 100 bar g, because with V-201 relieving at 110 % of set and C-201 at 100 % speed the discharge side is 11.3 bar higher. This also answers EQ-ep-03 (≈ 93 bar g before accumulation).	Add design P/T to every equipment tag on the next P&ID revision (R-201 100 bar g / 335 °C, V-201 88 bar g / 120 °C, …) and a spec break at the C-201 suction (zone A/B); confirm no block valves between C-201 discharge and V-201 (single PSV-201 protects both zones).	open
EQ-em-02	Mechanical	MEOH-V1-ME-DS-005 (E-203)	Simulator model / Design tab / MEOH-V1-PE-HMB-001	The model's final cooler is sensible-only (83.8 MW, CW 7740 t/h). With the condensing duty the real E-203 duty is 132.9 MW; at the model's CW flow the CW return would be about 46.8 °C, above the usual 45 °C limit for cooling-water return.	Datasheet E-203 is sized on the total 132.9 MW with 11433 t/h of CW (32 → 42 °C, assumed). Use this CW flow in the utility summary; if the model is revised, add the latent heat to the E-203 balance.	open
EQ-em-03	Mechanical	MEOH-V1-ME-EQL-001	MEOH-V1-PE-LSL-001	The line sizing list checks all loop pipe walls at 88 bar g with A106 Gr B at 260 °C and Class 600. Zone B lines (DN750-PG-202, DN750-PG-201, DN600-PG-213, DN900-PG-251, quench lines) are 100 bar g; DN600-PG-213 and DN900-PG-251 run up to 335 °C in hydrogen service (carbon steel outside the Nelson curve).	Re-check zone B wall thickness at 100 bar g and the design temperatures of this EQL; Cr-Mo (1¼Cr-½Mo or 2¼Cr-1Mo) for the converter inlet/outlet piping and Class 900 flanges (assumed) on zone B.	open

No.	Raised in	From	To	Issue	Proposed	Status
EQ-em-04	Mechanical	MEOH-V1-ME-DS-002 / DS-007	C&E MEOH-V1-IN-CE-001 / HAZOP MEOH-V1-SA-HZP-001	Total loss of cooling water to E-203 (model, trips as installed): TSHH-2xx any bed thermocouple above 300 °C trips the make-up at 488 s, but C-201 keeps circulating and the separator gas reaches about 351 °C (above 60 °C after 12 s); V-201 (120 °C) and C-201 (120 °C) are not designed for this.	Add a C-201 stop on high separator / C-201 suction temperature (e.g. TSHH-253 at 65 °C, assumed) to the C&E, or design V-201 and C-201 for the upset; review in the HAZOP follow-up.	open
EQ-em-05	Mechanical	MEOH-V1-ME-EQL-001	MEOH-V1-PE-RLB-001 (HV-205 depressuring)	The model depressures isothermally (separator temperature follows the cooling water), so it cannot give the minimum metal temperature reached when HV-205 blows the loop down. V-201 and E-203 are thick carbon steel.	Run a dynamic blowdown study (adiabatic gas expansion, wall heat transfer) and set the MDMT; −29 °C with impact testing is assumed on the datasheets until then.	open
EQ-er-01	Relief & vent	MEOH-V1-PE-RLB-001	MEOH-V1-PE-HMB-001 (battery limits) / front-end	The front end's maximum delivery pressure is not defined. In the model the make-up source is 87 bar a (≈ 86 bar g), so the loop can never reach the PSV-201 set (purge-shut run: 85.99 bar g). The relief loads S1–S3 assume a front end able to deliver 402 kNm³/h (110 % of design) at the relieving pressure, i.e. a shut-in / settle-out pressure of about 100 bar g (assumed).	Confirm the syngas compressor shut-in (settle-out) pressure and maximum capacity. If it cannot exceed 88 bar g, S1–S3 are not credible and the fire case S4 governs (smaller PSV); if it is higher, keep the governing case of this document.	open
EQ-er-02	Relief & vent	MEOH-V1-PE-RLB-001	MEOH-V1-IN-CE-001 / MEOH-V1-IN-SRS-001	PSHH-203 trips at 85 bar g = 96.6 % of the PSV-201 set. A conventional spring valve can simmer and leak from about 93–95 % of its set, i.e. before or at the trip; the operating margin between PAH 78, the trip and the set is only 3 bar.	Use a pilot-operated PSV-201 (tight to ~98 % of set), or lower PSHH-203 to ≤ 83 bar g (check spurious trips against PAH 78 bar g), or raise the zone A design / set pressure.	open
EQ-er-03	Relief & vent	MEOH-V1-PE-RLB-001	MEOH-PID-002 / Safety layers tab (TSV-203)	E-203 cooling-water side (10 bar g, assumed) is below 10/13 of the process side (100 bar g): a tube rupture needs relief on the CW side, 6342 kg/h of gas → 1575 mm² (L). Neither TSV-203 (thermal, Safety layers tab) nor any CW-side relief is drawn on MEOH-PID-002.	Add TSV-203 and a tube-rupture relief (PSV L or bursting disc to a safe location / flare) on the E-203 shell, or raise the CW-side design to ≥ 77 bar g; consider a dynamic tube-rupture check (liquid-full shell).	open
EQ-er-04	Relief & vent	MEOH-V1-PE-RLB-001	MEOH-PID-001	E-204 steam side (45 bar g, assumed) is below 10/13 of the process side: tube rupture sends 5509 kg/h of loop gas into the HP steam system; no steam-side relief is drawn.	Add a steam-side PSV on E-204 (H, set 45 bar g, to flare — it would relieve hydrogen-rich gas) or confirm the HP steam header relief covers it; keep HV-214 and the steam block valve closed outside start-up.	open
EQ-er-05	Relief & vent	MEOH-V1-PE-RLB-001	MEOH-PID-002 / HAZOP R03	HAZOP R03 asks for PSV-205 on the let-down drum sized for full gas blow-by; it is not on the P&ID. With LV-204 and LB-204 wide open the blow-by is about 35022 kg/h of loop gas; at a 10 bar g drum (assumed) this needs 8201 mm² (R).	Show PSV-205 (R) on the let-down drum; also apply HAZOP R02 (LSLL-204 closing a new XV-204) to reduce the demand.	open
EQ-er-06	Relief & vent	MEOH-V1-PE-RLB-001	MEOH-PID-002 note 6 / HAZOP R15	HV-205 restriction orifice is "TBC" on the P&ID. A 49 mm bore (C_d 0.61) reproduces the model blowdown rate (122.7 kNm³/h, 31909 kg/h at 75 bar g). With it the model reaches 50 % of the V-201 design in 5.1 min but 6.9 bar g only after 25.1 min, and HV-205 is a manual field valve.	Write the RO bore on the P&ID; make HV-205 a remotely operated, fail-open blowdown valve (HAZOP R15); confirm the flare capacity for the larger of PSV-201 and blowdown.	open
EQ-er-07	Relief & vent	MEOH-V1-PE-RLB-001	MEOH-PID-002	PSV-201 is a single valve with no isolation valves shown; it protects the whole loop, so it cannot be tested or replaced on line.	Install 2 × 100 % PSVs with interlocked (car-sealed open) inlet and outlet isolation valves, typical practice, and show them on the P&ID.	open

No.	Raised in	From	To	Issue	Proposed	Status
EQ-ei-01	Instrumentation & control	MEOH-V1-IN-INL-001	MEOH-PID-001	FI-214 (E-204 start-up steam flow, 0–60 t/h) is an indicator in the simulator/DCS but no flow element or bubble is drawn on the HP steam line to E-204 on MEOH-PID-001.	Add FE/FT-214 and FI-214 on the HP steam line upstream of HV-214 in MEOH-PID-001 rev B; keep it in the INL.	open
EQ-ei-02	Instrumentation & control	MEOH-V1-IN-INL-001	MEOH-V1-IN-CE-001	The E-204 steam valve is tagged HV-214 on MEOH-PID-001 (FC, driven by HIC-214), but the C&E effect column names the hand controller HIC-214 as the final element.	Name the effect "HV-214 (via HIC-214)" in the C&E rev R1 so the logic, the P&ID and the INL use one valve tag.	open
EQ-ei-03	Instrumentation & control	MEOH-V1-IN-INL-001	MEOH-PID-002 / MEOH-V1-IN-SRS-001	PSHH-203 and LSHH-204 have their own transmitters (SRS SIF-3, SIF-4) but no transmitter tag on MEOH-PID-002; the INL cannot list them as separate devices.	Assign SIS transmitter tags (e.g. a "B" suffix per site convention) on the P&ID and in the INL, with their own process connections.	open
EQ-ei-04	Instrumentation & control	MEOH-V1-IN-INL-001	Simulator model / MEOH-V1-IN-SRS-001	XV-202 is spring-return fail closed (SRS, P&ID note 2), but in the model an instrument-air failure moves only the control valves; XV-202 stays open. FV-202 (FC) still stops the make-up, so the safe state is reached by the BPCS valve, not the SIS valve.	Model XV-202 closing on instrument-air loss (or confirm the actuator design); keep FV-202 FC as the second barrier.	open
EQ-ei-05	Instrumentation & control	MEOH-V1-IN-INL-001	Simulator model (JI-201) / MEOH-V1-EE-LL-001	JI-201 range is 0–12 MW, but the model at 100 % C-201 speed absorbs about 12.6 MW: the indication would saturate at maximum circulation.	Range JI-201 0–16 MW (covers the 14 MW motor rating in MEOH-V1-EE-LL-001).	open
EQ-ei-06	Instrumentation & control	MEOH-V1-IN-INL-001	MEOH-V1-IN-CE-001 (PERM-3)	Start permissive PERM-3 needs every bed thermocouple below 280 °C, but the bed 2–4 exit high alarms are at 285 °C: a restart can be held off by a temperature that raises no alarm.	Review the TI-224/234/244 high alarm (e.g. 278 °C) or add a "PERM-3 not met" message on the SIS screen.	open
EQ-ei-07	Instrumentation & control	MEOH-V1-IN-INL-001	Simulator model (display ranges)	The model shows the bed thermocouples on a 150–320 °C scale and runs the TIC-2x1 PIDs on 100–300 °C, but a cold start-up begins at 30 °C and the trip is 300 °C.	Calibrate all converter thermocouples 0–400 °C (this INL); keep the DCS PID span 100–300 °C as a configuration item.	open
EQ-ei-08	Instrumentation & control	MEOH-V1-IN-INL-001	MEOH-V1-IN-SRS-001	FSLL-201 (SIS) and FIC-201 (BPCS) share FT-201, and TSHH-2xx shares the 12 indicating thermocouples: an FT-201 failure is both an initiating cause and a disabled protection layer.	Follow the SRS proposals (FSLL-201BX on its own FT-201B, TSHH-2oo3 on dedicated elements); add the new transmitters to the INL and the I/O count.	open
EQ-ee-01	Electrical	MEOH-V1-EE-LL-001	MEOH-PID-002 note 4 (C-201 vendor)	C-201 seal and lube-oil systems are "per vendor": the lube-oil pumps, heaters, seal-gas and VSD auxiliary loads in the load list are assumed.	Obtain the C-201 package auxiliary load list; show the lube-oil console and the seal-gas panel on the P&ID.	open
EQ-ee-02	Electrical	MEOH-V1-EE-HAC-001	MEOH-PID-002 (E-203) / HAZOP	An E-203 tube leak carries hydrogen-rich gas at about 75 bar g into the cooling-water return, which degasses at the cooling tower outside the unit: a release source the unit HAC does not cover.	Add gas detection (or a degassing pot with vent to a safe location) on the E-203 CW return; extend the classification to the cooling-tower return header.	open
EQ-ee-03	Electrical	MEOH-V1-EE-LL-001	MEOH-PID-001	The AIT-208 sample (converter outlet, CH3OH 4.78 % at 78.39 bar a) condenses below about 101.9 °C; no heat tracing is shown on the analyser sample lines.	Show electric heat tracing on the AIT-205/208 sample lines (maintain ≥ 120 °C on AIT-208) on MEOH-PID-001 rev B; keep the load in the load list.	open
EQ-ee-04	Electrical	MEOH-V1-EE-HAC-001	Downstream let-down drum (off-sheet)	LV-204 lets crude methanol down to the let-down drum (about 6 bar g, off-sheet), where dissolved H2 and CO2 flash with methanol vapour: that drum and its vent are not classified here.	Classify the let-down drum, its flash-gas line and the crude-methanol tankage with the downstream unit (HAC rev 1).	open

No.	Raised in	From	To	Issue	Proposed	Status
EQ-ee-05	Electrical	MEOH-V1-EE-LL-001	Simulator model (C-201 curve) / MEOH-PID-002	The motor rating (14 MW) is set by the model's 100 % speed point (12.59 MW absorbed at 2415 kNm³/h); the model has no maximum continuous speed or surge limit and the P&ID gives no driver rating.	Confirm the C-201 maximum continuous speed and the rated point with the compressor curve; show the driver rating on MEOH-PID-002.	open
EQ-ec-01	Procurement to commissioning	MEOH-V1-PM-COM-001	MEOH-V1-ME-EQL-001	Test pressures are computed from design pressures derived here by the typical rule (1.1 × maximum operating, and not less than PSV-201 set 88 bar g plus the loop pressure drop from the item to V-201): 94 bar g at C-201 discharge down to 88 bar g at V-201. The equipment list is not issued yet.	Confirm the design pressures and temperatures in the EQL and the datasheets; re-issue the pressure-test table. A single loop design pressure (the highest value) would simplify the piping class.	answered by MEOH-V1-ME-EQL-001 EQL zones: A (V-201, C-201 suction, E-203 outlet piping) 88 bar g = PSV-201 set; B (C-201 casing, E-202 both sides, E-201 both sides, E-204 tube side, R-201, E-203 tube side) 100 bar g (ASME UG-125: every item protected …
EQ-ec-02	Procurement to commissioning	MEOH-V1-PM-PKG-001	R-201 datasheet / model header	Catalyst quantity: 184 m³ × 1150 kg/m³ = 211.6 t (model header), but the bed masses K.Wb add up to 208.3 t (also on MEOH-PID-001). The equipment list uses a loaded bulk density of 1131 kg/m³ (184.1 m³ → 208.3 t), consistent with K.Wb; the 1150 kg/m³ in the model header is not.	Order on volume (184 m³ plus a spare, assumed 5 %) with the loaded density confirmed by the catalyst supplier; correct one of the two numbers.	open
EQ-ec-03	Procurement to commissioning	MEOH-V1-PM-PKG-001	C-201 datasheet (Mechanical tab)	C-201 is sized for loop gas of MW 5.83 kg/kmol. Dry-out and catalyst heat-up on nitrogen (MW 28) give about 4.8 × the gas density, so head and power at the same speed and pressure would overload the drive.	Add a nitrogen running case to the datasheet (reduced pressure about 1/4.8 of the loop pressure or reduced speed) and check the motor and drive before the PO.	open
EQ-ec-04	Procurement to commissioning	MEOH-V1-PM-PKG-001	MEOH-V1-IN-SRS-001	FSLL-201 shares FT-201 with FIC-201 and TSHH-2xx uses the indicating bed thermocouples (HAZOP R01, SRS proposals FSLL-201B and TSHH 2oo3). Extra elements change the R-201 nozzle schedule, which is frozen at the R-201 PO (critical path).	Decide on the SRS proposals before the R-201 requisition; include FT-201B and the dedicated thermocouple nozzles as priced options.	open
EQ-ec-05	Procurement to commissioning	MEOH-V1-PM-COM-001	MEOH-PID-002	The hand valves LB-204, PB-203 and BV-231 used in the SOPs and on the simulator's Field tab are not on the EPC P&IDs, so they are missing from the piping bulk and the stroke/line-up lists.	Add them to MEOH-PID-002 (LV-204 and PV-203 bypasses, BV-231) or remove them from the SOPs.	open
EQ-ec-06	Procurement to commissioning	MEOH-V1-PM-COM-001	MEOH-V1-PE-RLB-001	Hydrotest pressures (e.g. 114.4 bar g on V-201) exceed the PSV-201 set 88 bar g; the HV-205 restriction orifice has no size on the P&ID; E-203 CW side (10 bar g assumed) and E-204 steam side (45 bar g) are far below the loop pressure (tube-rupture case).	PSV-201 removed or blinded during the hydrotest and re-installed with its certificate; size the HV-205 orifice and settle the tube-rupture protection of E-203 and E-204 in the relief load basis before ordering PKG-05 and PKG-10.	open
EQ-ec-07	Procurement to commissioning	MEOH-V1-PM-COM-001	SOP tab (SOP-MEOH-01)	SOP-MEOH-01 assumes catalyst already reduced by the supplier procedure; the commissioning plan needs a decision between in-situ reduction (H_2 in N_2 circulation, low-pressure C-201 case, water drained at V-201) and pre-reduced catalyst.	Decide the catalyst form at the PKG-02 requisition; write the reduction SOP if in-situ.	open

No.	Raised in	From	To	Issue	Proposed	Status
EQ-ly-01	Layout & isometrics	MEOH-V1-LAY-001	MEOH-PID-001 / MEOH-PID-002 · MEOH-V1-PE-LSL-001	The layout routes 28 lines but only 11 carry a P&ID line number. Two of the six isometrics (DN300-FL-X08, DN1200-CW-X11) and 15 other lines use provisional numbers: the 8 unnumbered process lines of EQ-ep-06 (bed-1 branch, TV-211 bypass, E-201 → E-204, E-201 → E-202, E-202 → E-203, E-203A → B, E-203 → V-201, the PSV-201 and HV-205 flare lines) and every utility line (flare sub-header, CW supply and return, HP steam and condensate to E-204, N_2, instrument air, V-201 drain), which the P&IDs show only as arrows.	Number every line on MEOH-PID-001/002 rev B with the next free numbers of each service (PG-2xx, FL-2xx, CW-2xx, HS-2xx …), add the utility lines with their sizes to the line sizing list, and reissue the isometrics with the final numbers.	open
EQ-ly-02	Layout & isometrics	MEOH-V1-LAY-001	MEOH-PID-002 / C-201 vendor (DS-007)	DN900-PG-205 rises from the V-201 top (EL +9.99), runs on the rack at EL +6.5 and drops 3.1 m to the top suction nozzle of the grade-mounted C-201 (EL +3.40): the low point of the suction line is the machine. The loop gas leaves V-201 saturated at 40 °C, so condensate (cooler ambient, a demister upset) runs to C-201; V-201 is the only knock-out drum and LSHH-204 protects only the V-201 level. The layout fits a low-point drain only.	Add a suction drain pot (or a small knock-out drum) at the C-201 inlet with a level gauge, LSH alarm and drain to the closed drain; insulate PG-205; or mount C-201 on a table-top so the suction line falls back to V-201 (1:500). Confirm the vendor's liquid tolerance.	open
EQ-ly-03	Layout & isometrics	MEOH-V1-LAY-001	Structural / platform layout (E-201)	The E-201 top platform PF-E1 (EL +16.8) is reached only over the bridge BR-E1 from the R-201 bed 2 platform: from its far corner an operator walks 9.4 m before two ways down are available, above the 7.5 m dead-end limit used here (assumed). A fire at the bridge or the quench risers would trap him.	Add a caged ladder from PF-E1 down the west side of E-201 with a rest landing on the STR-E201 structure (EL +10.0), or extend the structure as a second stair; recheck when the E-201 structure is detailed.	open
EQ-ly-04	Layout & isometrics	MEOH-V1-LAY-001	MEOH-PID-002 note 6 · EQ-er-06 · HAZOP R15	HV-205 is a manual blowdown valve. Its line must fall continuously to the flare sub-header (EL +9), so the valve sits high on the V-201 top platform PF-V1 (EL +9.2), reached only by two ladder flights. In the emergency HV-205 is meant for (fire, leak) the operator would have to climb onto the top of the loop inventory vessel.	Make HV-205 a remotely operated, fail-open blowdown valve (BDV) with a hand switch in the CCR and at a safe location at grade (as EQ-er-06 / HAZOP R15 propose); keep a local hand-wheel only for maintenance. If it stays manual, move it to a stair-served platform on the rack side with the line still free-draining.	open
EQ-ly-05	Layout & isometrics	MEOH-V1-LAY-001	MEOH-PID-002 rev A · EQ-ec-05	The layout places 8 local gauges an outside operator reads (PG-202, PG-261, PG-262, PG-203, LG-204, TG-263, TG-264, FG-203), 16 hand valves (BV-202 … BV-292, from the field model) and the battery-limit valves; none is drawn on MEOH-PID-001/002 rev A, and their ranges, sizes and car-seal status exist only in the layout and the simulator.	Show every local gauge (with range), every block, bypass, drain and isolation valve (with NO / CSO / LC) and the BL valves with spectacle blinds on rev B; add them to the piping bulk and the valve list (see EQ-ec-05).	open
EQ-ej-01	Project & deliverables	MEOH-V1-PM-DOC-001	PFD-201 / MEOH-PID-001 / MEOH-PID-002	4 of the 24 documents do not follow the unit numbering MEOH-V1-<discipline>-<type>-<nnn>: PFD-201, MEOH-PID-001, MEOH-PID-002, MEOH-V1-LAY-001. PFD-201 and the P&IDs were drawn before the EPC set; MEOH-V1-LAY-001 has no discipline code. Searching or filing by discipline misses them.	Keep the drawing numbers on the sheets and add register aliases (e.g. MEOH-V1-PE-PFD-001, MEOH-V1-PE-PID-001/002, MEOH-V1-PL-LAY-001, assumed codes) at the next revision; or renumber at rev B.	open
No.	**Raised in**	**From**	**To**	**Issue**	**Proposed**	**Status**
EQ-ej-02	Project & deliverables	MEOH-V1-PM-DOC-001	MEOH-V1-IN-CE-001 / MEOH-V1-IN-SRS-001	Three revision codes are in use (0, A, R0): the C&E and the SRS write "R0", the HAZOP report and the P&IDs "A", the	One revision scheme for the unit: A, B … for review and approval issues, 0, 1 … from the	open

No.	Raised in	From	To	Issue	Proposed	Status
			/ MEOH-V1-SA-HZP-001	EPC documents "0". The same status ("Issued for teaching") carries different codes. MEOH-V1-LAY-001 is issued as Rev 0 but its cover calls the plot plan "drawing rev A".	first issue for construction (here: for teaching). Re-issue the C&E and the SRS as Rev 0.	
EQ-ej-03	Project & deliverables	MEOH-V1-PM-DOC-001	MEOH-V1-IN-SRS-001 (SRS of SIF tab)	The SRS has a document number and is cited by the C&E, the instrument list and the package list, but no issued document exists: it lives only on the SRS of SIF tab (and its CSV). An unissued document cannot be revision-controlled or transmitted to the SIS vendor.	Issue MEOH-V1-IN-SRS-001 Rev 0 as a PDF from the SRS tab data before the SIS package (PKG) requisition.	open
EQ-ej-04	Project & deliverables	MEOH-V1-PM-DOC-001	MEOH-PID-001 / MEOH-PID-002	The P&IDs are at rev A (issued for review) while 18 documents built on them are at Rev 0. 22 of the 44 engineering queries ask for a P&ID change, none closed on the drawings yet. The simulator now models 16 hand valves (BV-202 … BV-292) that rev A does not show (EQ-ly-05, EQ-ec-05).	Issue MEOH-PID-001/002 rev B incorporating the accepted queries (line sizes, line numbers, missing instruments and local gauges, relief devices, hand valves), then re-check the LSL, INL, RLB and the layout MEOH-V1-LAY-001 against rev B.	open

5 Progress measurement (worked, weights assumed)

Engineering progress on an EPC project is measured, not estimated: each deliverable has a weight (its budget hours) and earns a fixed percentage at each revision milestone (the "rule of credit"). Progress = Σ (weight × credit) / Σ weight. Hours per document and the rule-of-credit percentages are typical EPC practice for a unit of this size, assumed for teaching; a real project takes them from its own estimate and progress-measurement procedure.

Milestone	Credit, %
Not started	0
Work in progress (internal draft)	30
Issued for review (Rev A)	60
Issued for approval (Rev B)	80
Issued for construction (Rev 0, here for teaching)	95
As-built / final	100

Document	Rev	Milestone	Hours (assumed)	Credit, %	Earned hours
MEOH-V1-PM-DOC-001	0	Issued for construction (Rev 0, here for teaching)	40	95	38
PFD-201	0	Issued for construction (Rev 0, here for teaching)	160	95	152
MEOH-PID-001	A	Issued for review (Rev A)	320	60	192
MEOH-PID-002	A	Issued for review (Rev A)	320	60	192
MEOH-V1-PE-HMB-001	0	Issued for construction (Rev 0, here for teaching)	400	95	380
MEOH-V1-PE-LSL-001	0	Issued for construction (Rev 0, here for teaching)	160	95	152
MEOH-V1-PE-RLB-001	0	Issued for construction (Rev 0, here for teaching)	240	95	228
MEOH-V1-ME-EQL-001	0	Issued for construction (Rev 0, here for teaching)	120	95	114
MEOH-V1-ME-DS-001	0	Issued for construction (Rev 0, here for teaching)	60	95	57
MEOH-V1-ME-DS-002	0	Issued for construction (Rev 0, here for teaching)	60	95	57
MEOH-V1-ME-DS-003	0	Issued for construction (Rev 0, here for teaching)	60	95	57
MEOH-V1-ME-DS-004	0	Issued for construction (Rev 0, here for teaching)	60	95	57
MEOH-V1-ME-DS-005	0	Issued for construction (Rev 0, here for teaching)	60	95	57
MEOH-V1-ME-DS-006	0	Issued for construction (Rev 0, here for teaching)	60	95	57
MEOH-V1-ME-DS-007	0	Issued for construction (Rev 0, here for teaching)	60	95	57
MEOH-V1-LAY-001	0	Issued for construction (Rev 0, here for teaching)	480	95	456
MEOH-V1-IN-INL-001	0	Issued for construction (Rev 0, here for teaching)	240	95	228
MEOH-V1-IN-CE-001	R0	Issued for construction (Rev 0, here for teaching)	120	95	114
MEOH-V1-IN-SRS-001	R0	Work in progress (internal draft)	200	30	60
MEOH-V1-EE-LL-001	0	Issued for construction (Rev 0, here for teaching)	80	95	76
MEOH-V1-EE-HAC-001	0	Issued for construction (Rev 0, here for teaching)	100	95	95
MEOH-V1-SA-HZP-001	A	Issued for review (Rev A)	400	60	240
MEOH-V1-PM-PKG-001	0	Issued for construction (Rev 0, here for teaching)	160	95	152

Document	Rev	Milestone	Hours (assumed)	Credit, %	Earned hours
MEOH-V1-PM-COM-001	0	Issued for construction (Rev 0, here for teaching)	240	95	228
Total			**4200**		**3496**

Worked: earned 3496 h of a budget of 4200 h → progress **83.2 %**. The SRS counts as work in progress (30 %) because it has a number but no issued document (EQ-ej-03); the P&IDs and the HAZOP report at rev A earn 60 %. Raising the two P&ID sheets to rev B would add 128 h (3.0 points).

6 Checks at issue

Computed by the data script (epc/eproj_data.js) on 2026-10-27 from the PDFs, the other tabs' data and the model. The OTS tab recomputes them in the page.

Check	Value	Result
Every PDF in epc/pdf is in the register	20 files	✓ (asserted by the data script)
Every issued PDF opens and has pages; size < 3 MB	largest 2.18 MB	✓
Every query's "from" document is in the register	48 queries	✓
PSV-201 set: RLB = EQL zone A = PFD-201 note 4	88 / 88 / 88 bar g	✓
PSV-201 set above the PSHH-203 trip (model K.tripP)	88 > 85 bar g	✓
C-201 power at design: model JI-201 = HMB = C-201 datasheet = load list	6528 / 6528 / 6528 / 6528 kW	✓
Every document dated and with a revision	24 documents	✓
Hand-overs cited by number in the receiving PDF	30 of 40	information
Open engineering queries	46 of 48	information